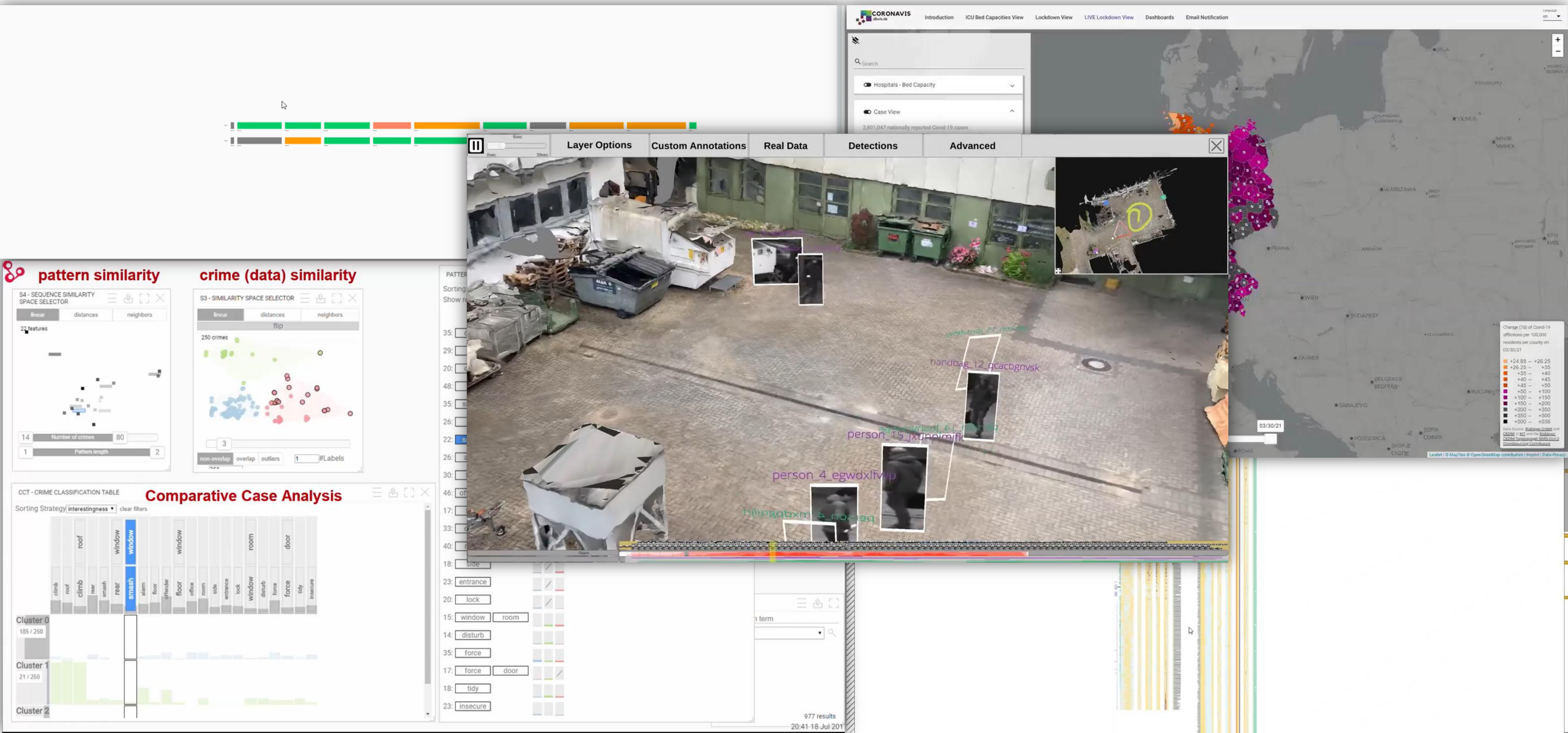
Visualization Foundations for Public Health

June 10th 2026

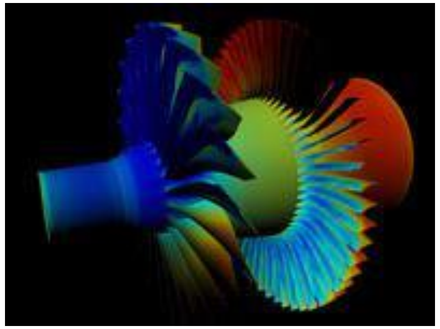
THE UNIVERSITY OF ARIZONA
Responsible Artificial Intelligence

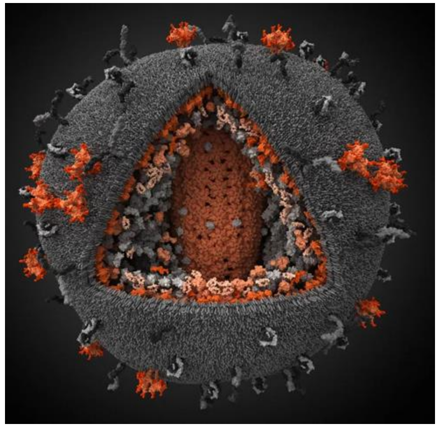
Wolfgang Jentner

Arizona Institute for AI and Society: AI^2S



Scientific Visualization & Information Visualization





Vis: Exploratory vs. Explanatory


"All the News That's Fit to Print"

The New York Times

Late Edition

NEW YORK, FRIDAY, MARCH 27, 2020

JOB LOSSES SOAR; U.S. VIRUS CASES TOP WORLD

New Data Shows Staggering Toll of Outbreak

Under Trump, Unfilled Posts Hinder Action

Online Class With No Way To Get There

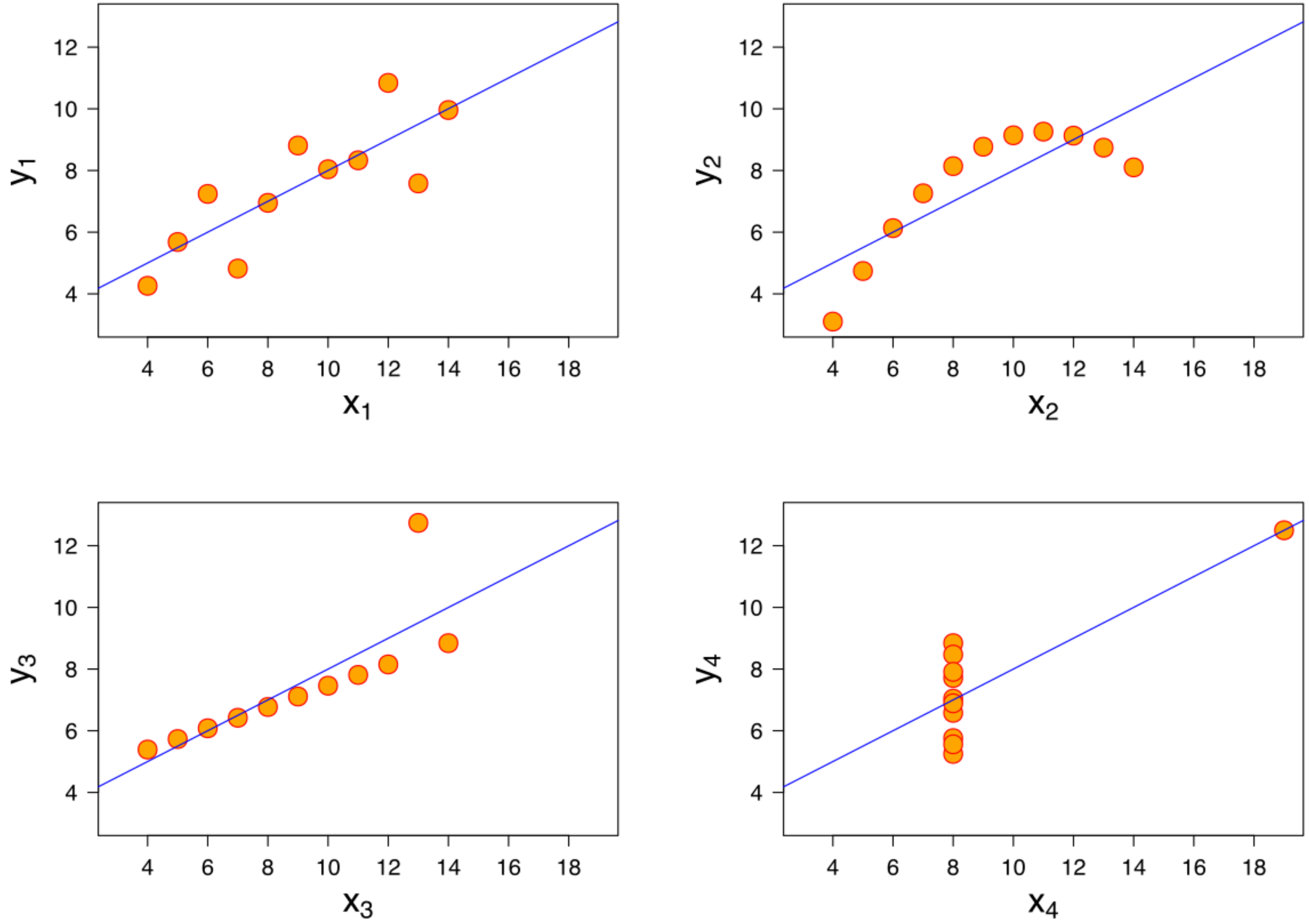
Courage at a Brooklyn Hospital, At the Front of an Invisible War

With Supplies Waning, Workers Soldier On

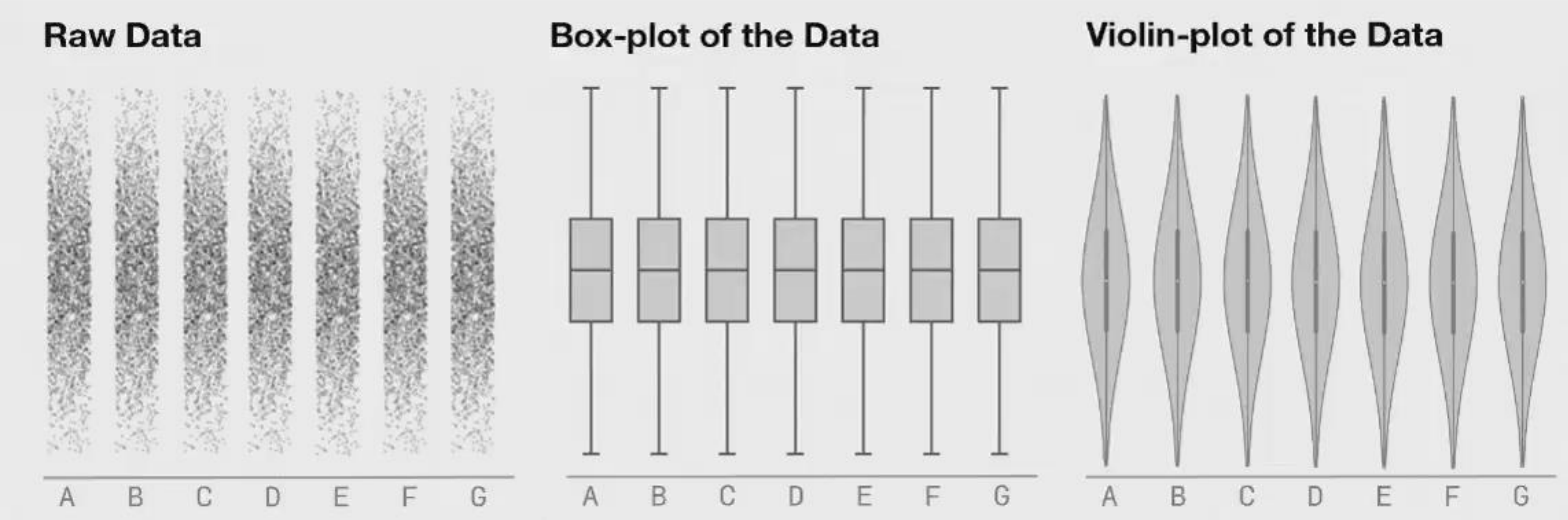
No Crowd, but I'll Take You Out to the Ballgame

A Fan Writes a Fantasy for Opening Day

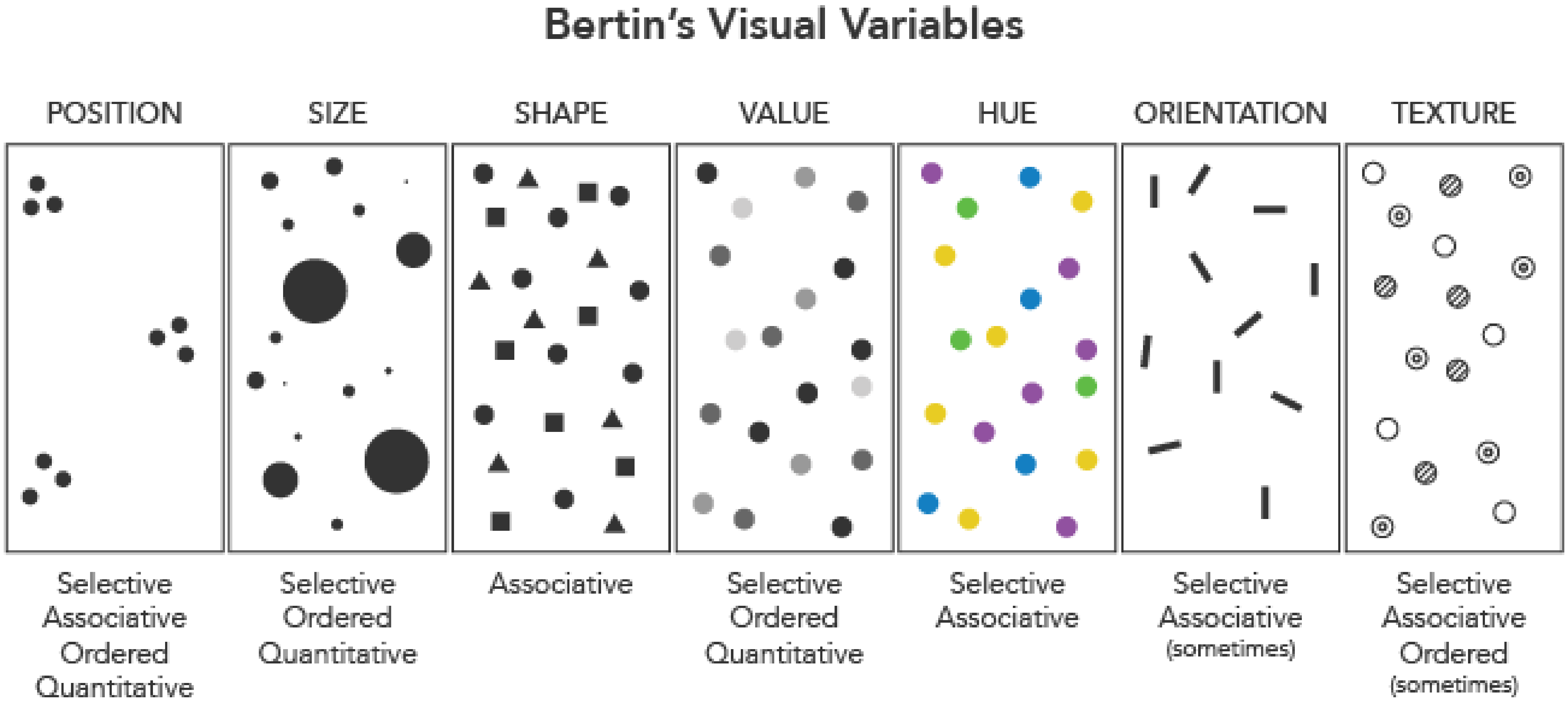
Why visualize?



Why visualize?



Visualize how?



Jaques Bertin, 1974: **Semiology of graphics**

Encoder -- Decoder

Designer:

Encoding Data & Features into visual variables

Human / User:

Decoding Data from visual stimuli

Mapping Features to Visual Variables

Longitude	Latitude	Temperature	Population over 10M?	Founded before 1700?
-118.2681861	34.0378592	90	Yes	No
-122.446747	37.733795	80	No	No
-75.000000	43.000000	95	Yes	Yes
-71.057083	42.361145	75	No	Yes

Longitude, Latitude → Mercator Projection → x, y

Temperature → Area, Linear Normalization → Area Size

Founded → Shape, Yes = Triangle, No = Circle

Population → Color, Discrete, Yes = Orange, No = Blue

Combine any visual variables?

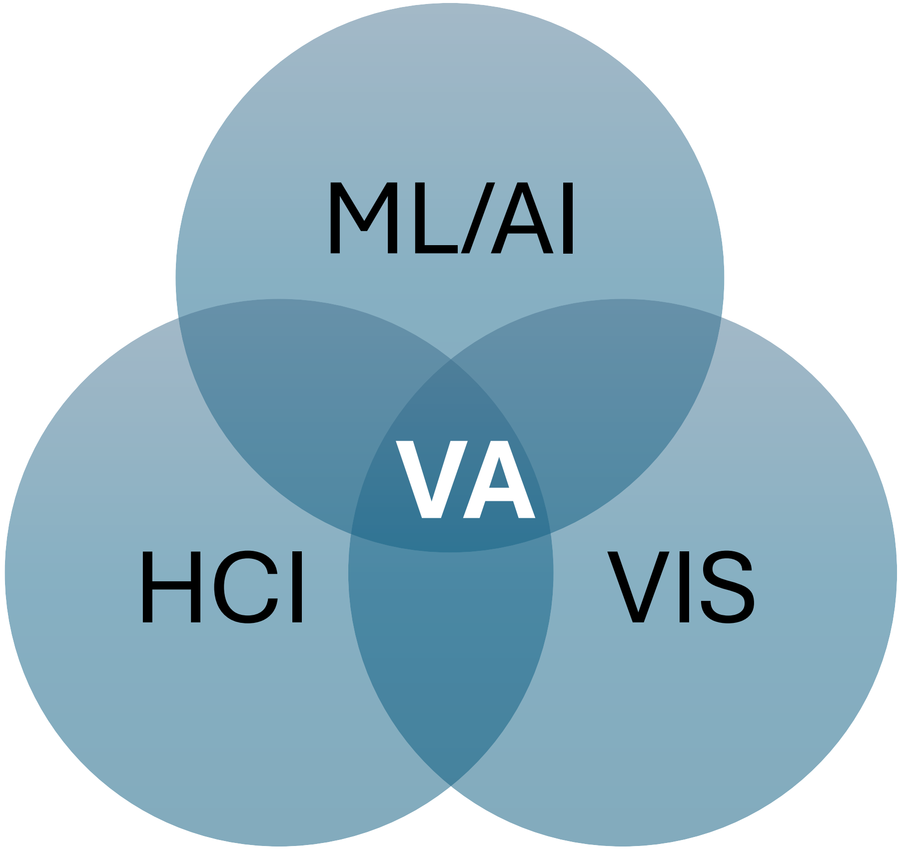
Variables:

Position (x & y)
Shape
Area
Color

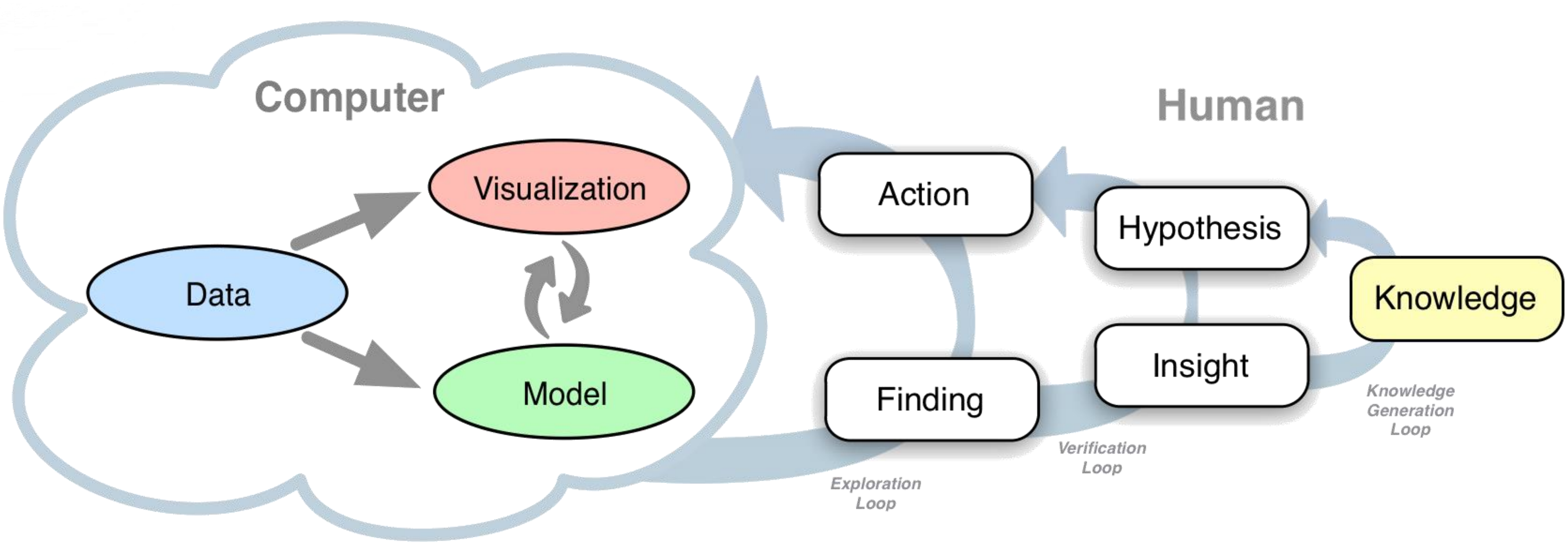
Visual Analytics

Visual Analytics combines human (expert-) knowledge with artificial intelligence and machine learning through interactive visual interfaces and user interfaces.

Visual Analytics



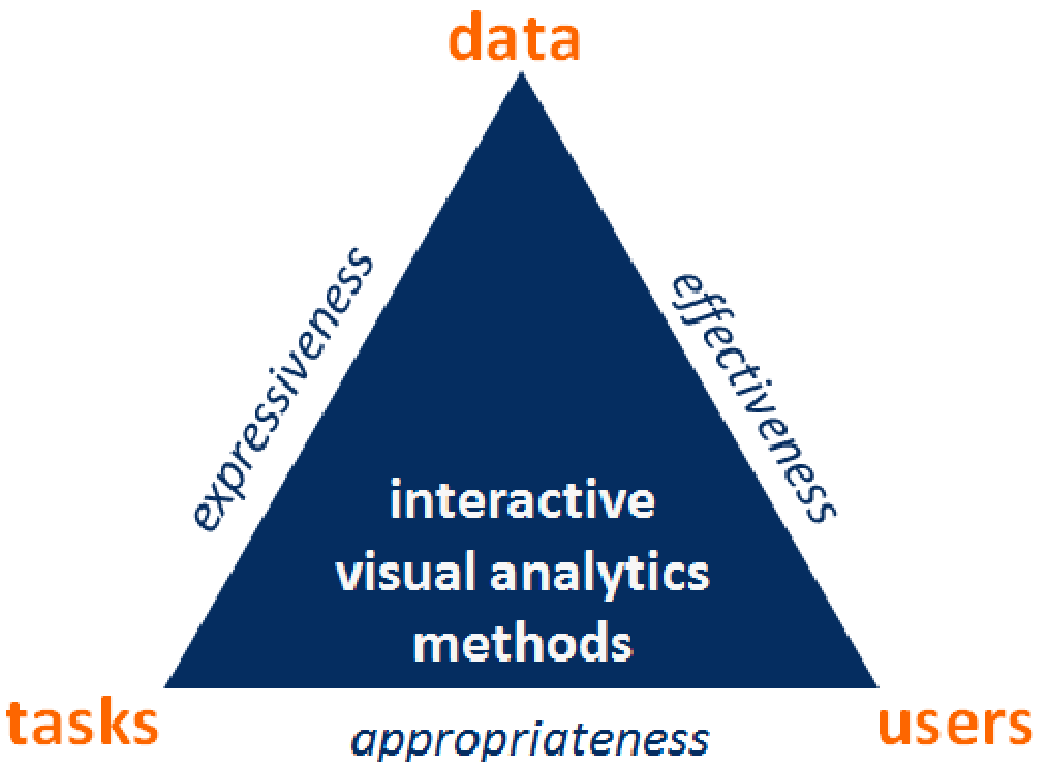
Knowledge Generation Model for Visual Analytics



Dominik Sacha, Andreas Stoffel, Florian Stoffel, Bum Chul Kwon, Geoffrey P. Ellis, Daniel A. Keim:
Knowledge Generation Model for Visual Analytics. IEEE Trans. Vis. Comput. Graph. 20(12): 1604-1613 (2014)

Design Considerations

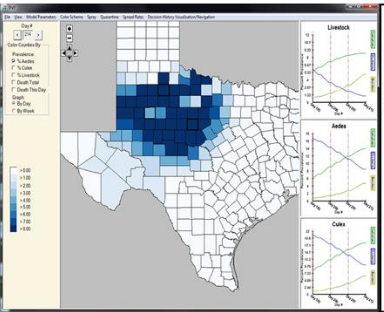
Plan ahead!

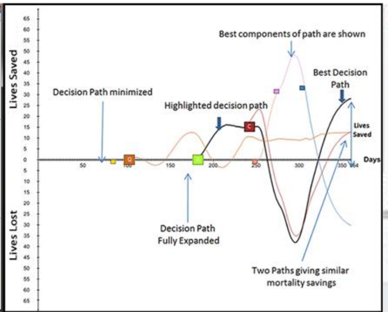


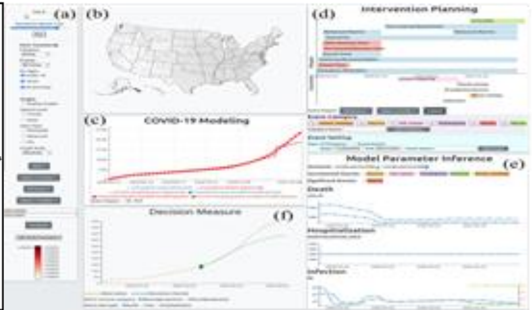
S. Miksch, W. Aigner: **A matter of time: Applying a data-users-tasks design triangle to visual analytics of time-oriented data.** Computers & graphics 1 February 2014. 10.1016/j.cag.2013.11.002

Goal: Improved Public Health Decision Making, Planning & Response

- Enable earlier detection
- Enable more effective decision-making through innovative interactive visualization, analysis, and decision-making tools
  - Provide the right information, in the right format, within the right time to solve the problem
  - Turn data deluge into a pool of relevant, actionable knowledge
  - Enable the user to be more effective from planning to detection to response to recovery
  - Enable effective communication of information

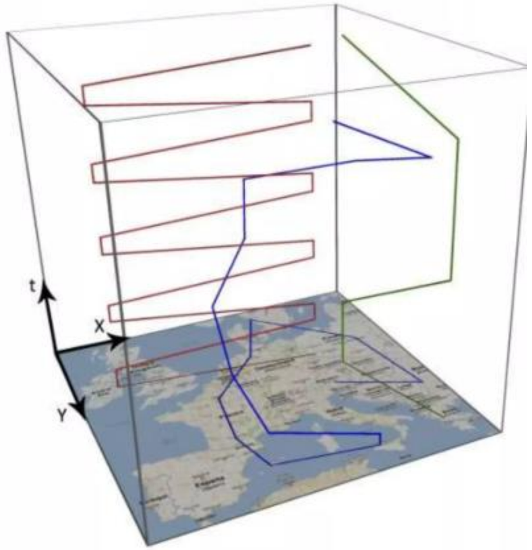




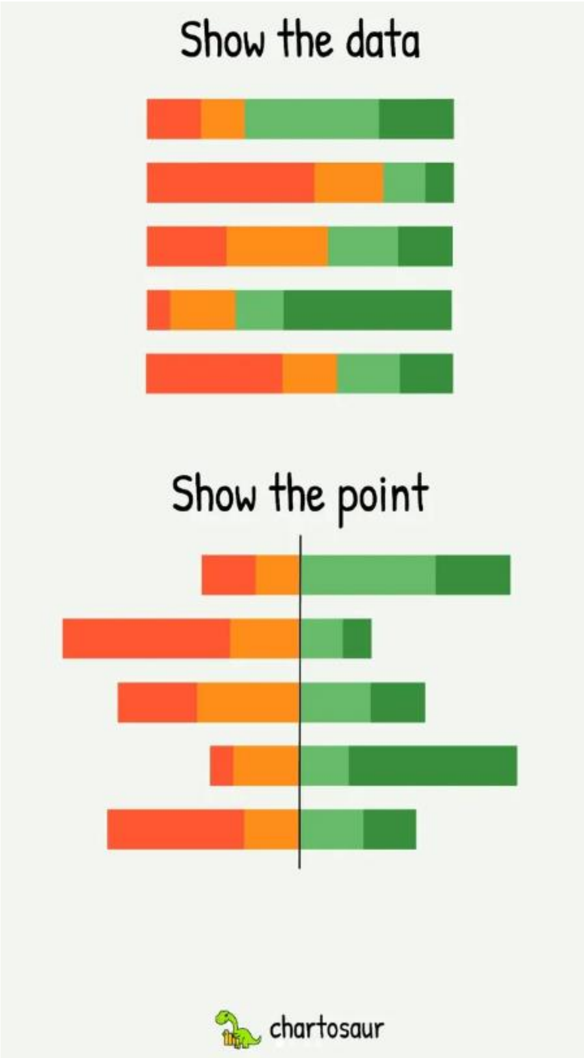


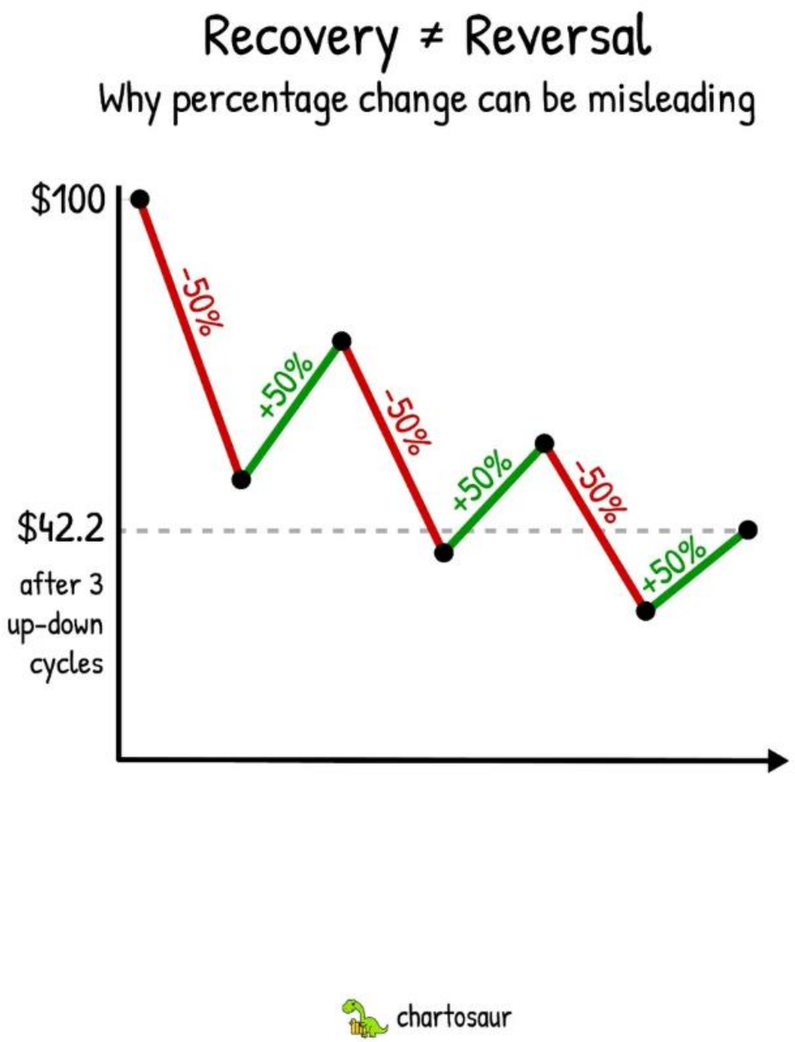
Public Health is complex!

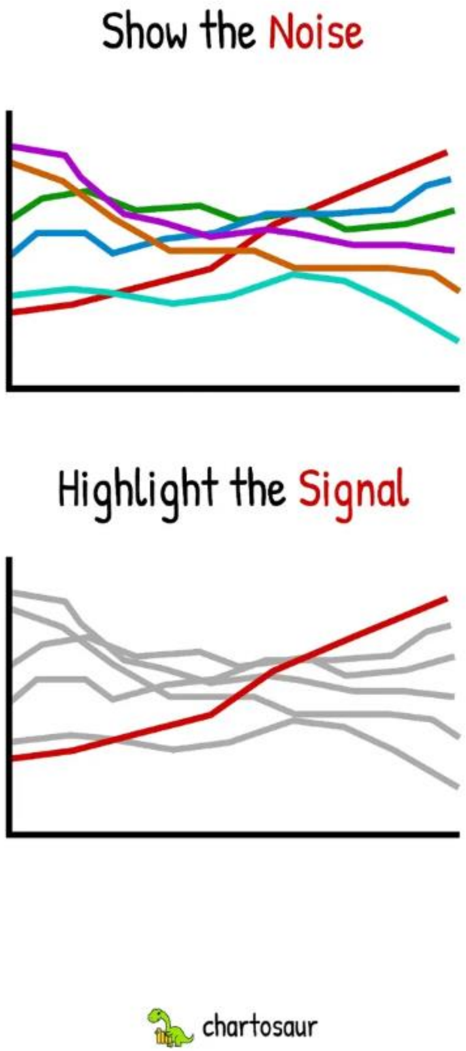
- Multiple data sources
- Massive amounts of data
- Quality issues (uncertainty)
- Spatio-temporal data
  - needs three or four dimensions ➔ only 2D screen space available



Design with purpose!

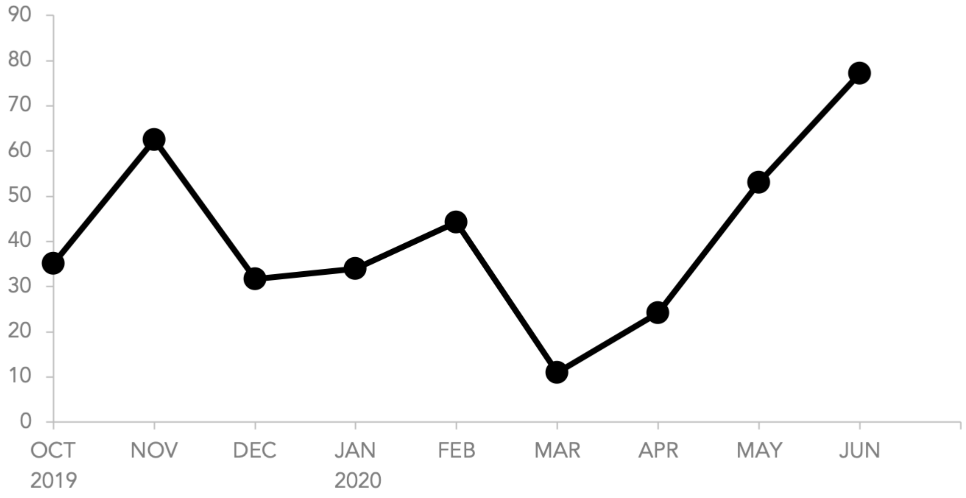




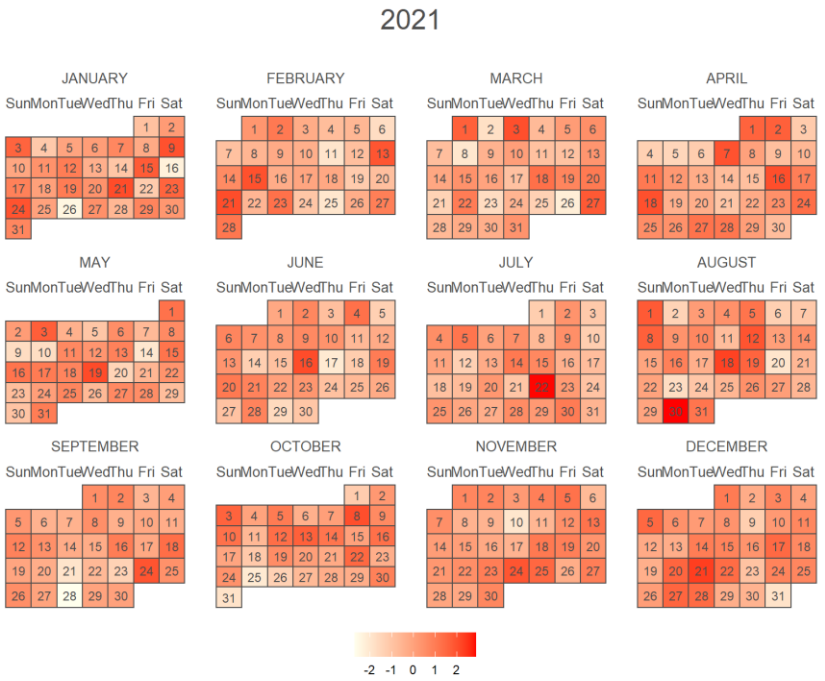


Designing Temporal Charts

Choices

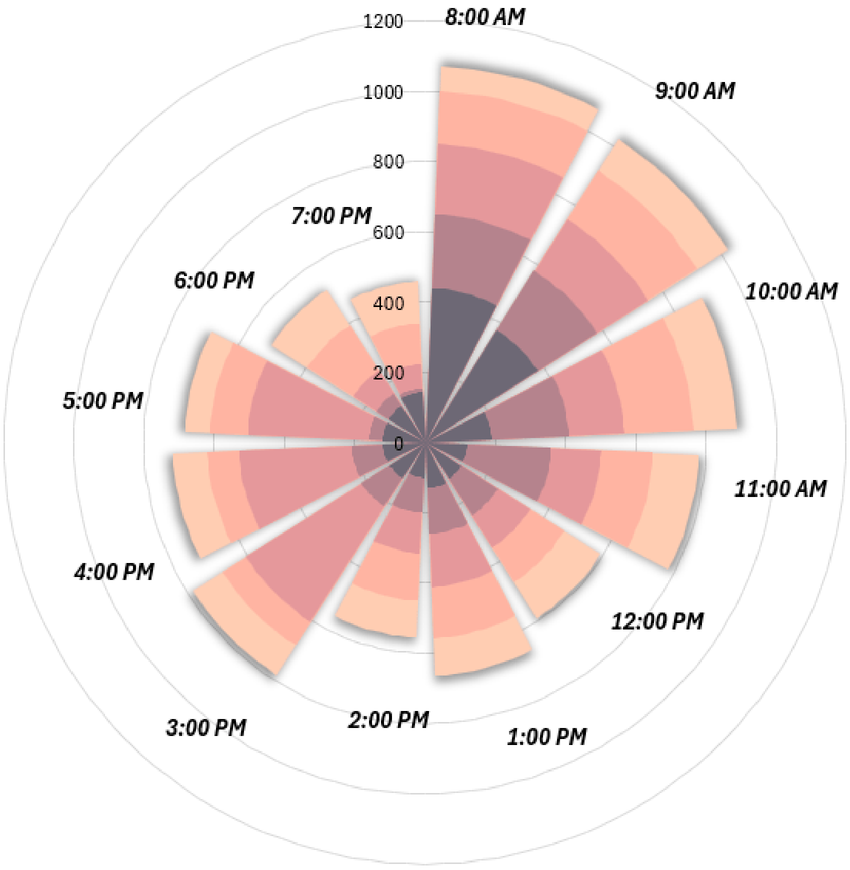


90
80
70
60
50
40
30
20
10
0
OCT 2019
NOV
DEC
JAN 2020
FEB
MAR
APR
MAY
JUN

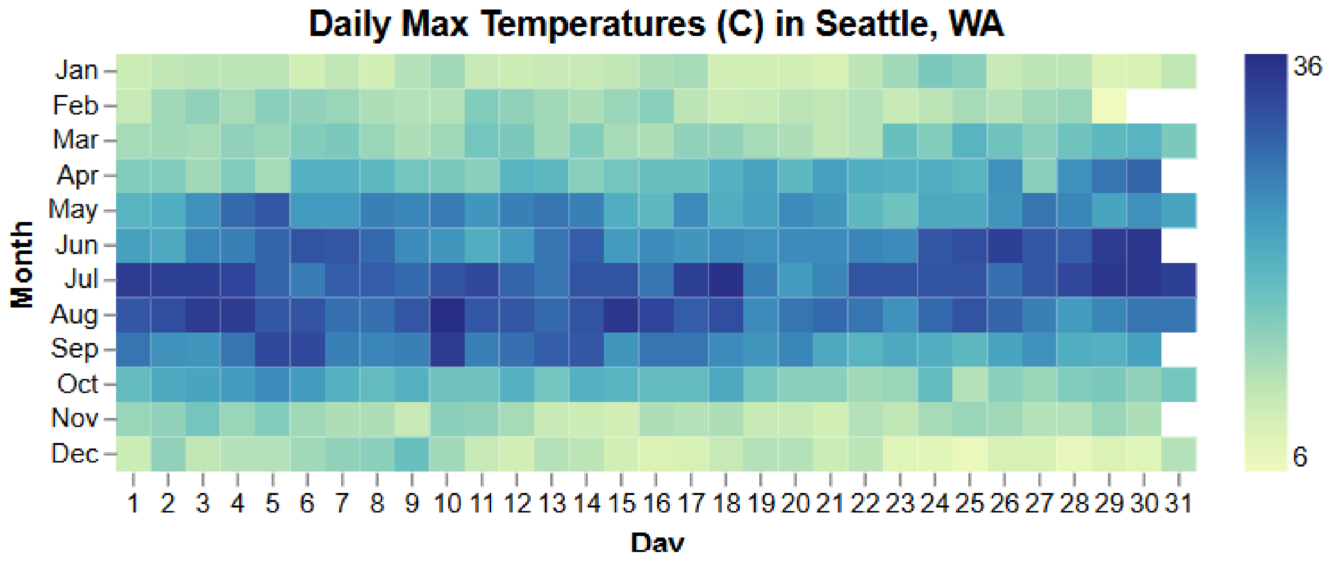


2021
JANUARY
FEBRUARY
MARCH
APRIL
MAY
JUNE
JULY
AUGUST
SEPTEMBER
OCTOBER
NOVEMBER
DECEMBER
Sun Mon Tue Wed Thu Fri Sat
-2 -1 0 1 2

Choices

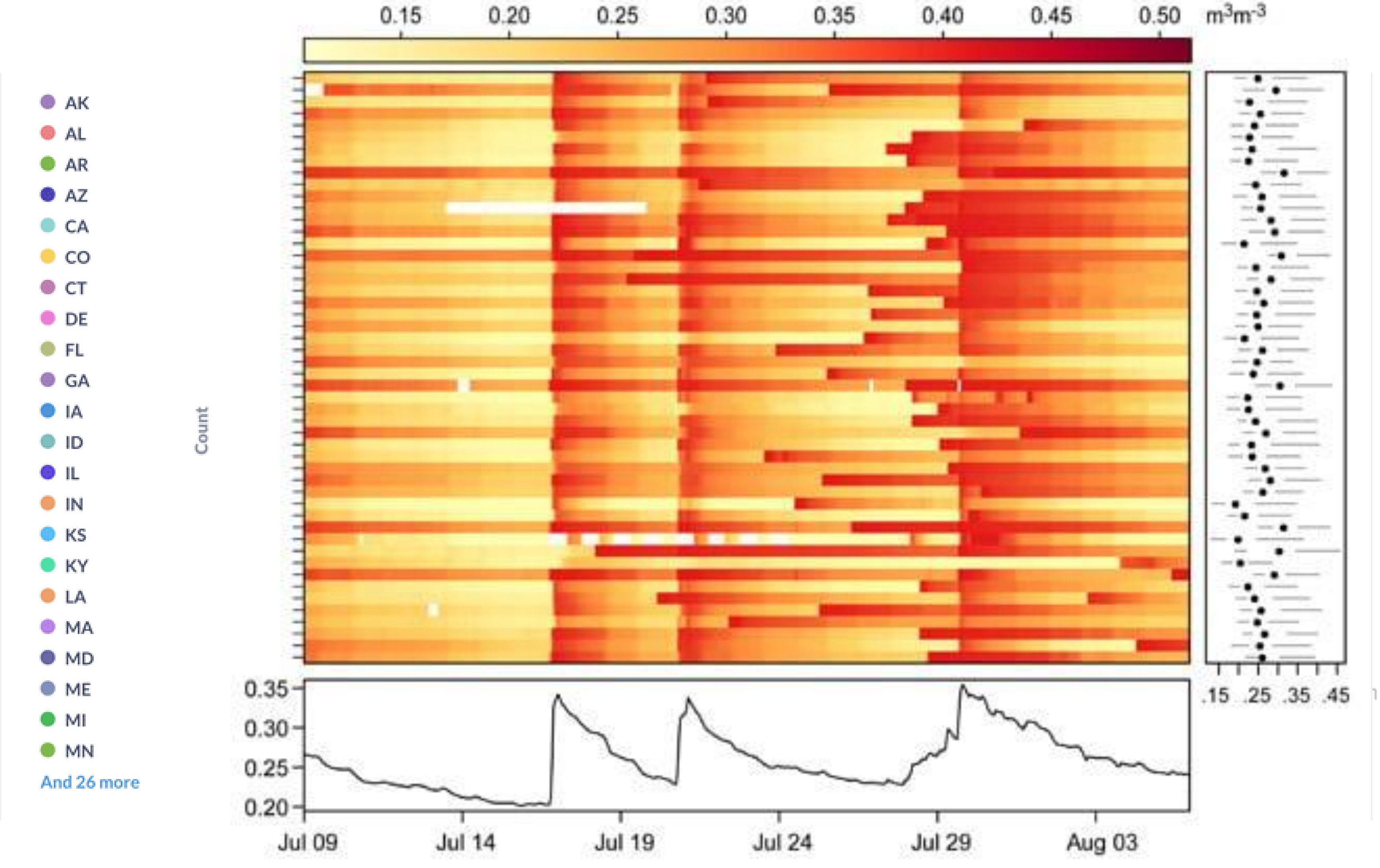


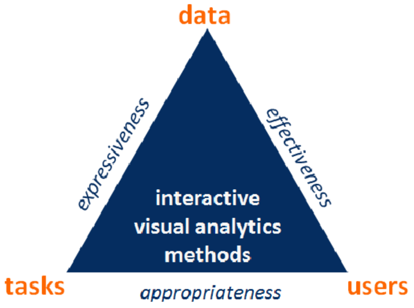
8:00 AM
9:00 AM
10:00 AM
11:00 AM
12:00 PM
1:00 PM
2:00 PM
3:00 PM
4:00 PM
5:00 PM
6:00 PM
7:00 PM
1200
1000
800
600
400
200
0



Daily Max Temperatures (C) in Seattle, WA
Month
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
Dav
36
6

Beware of overplotting!

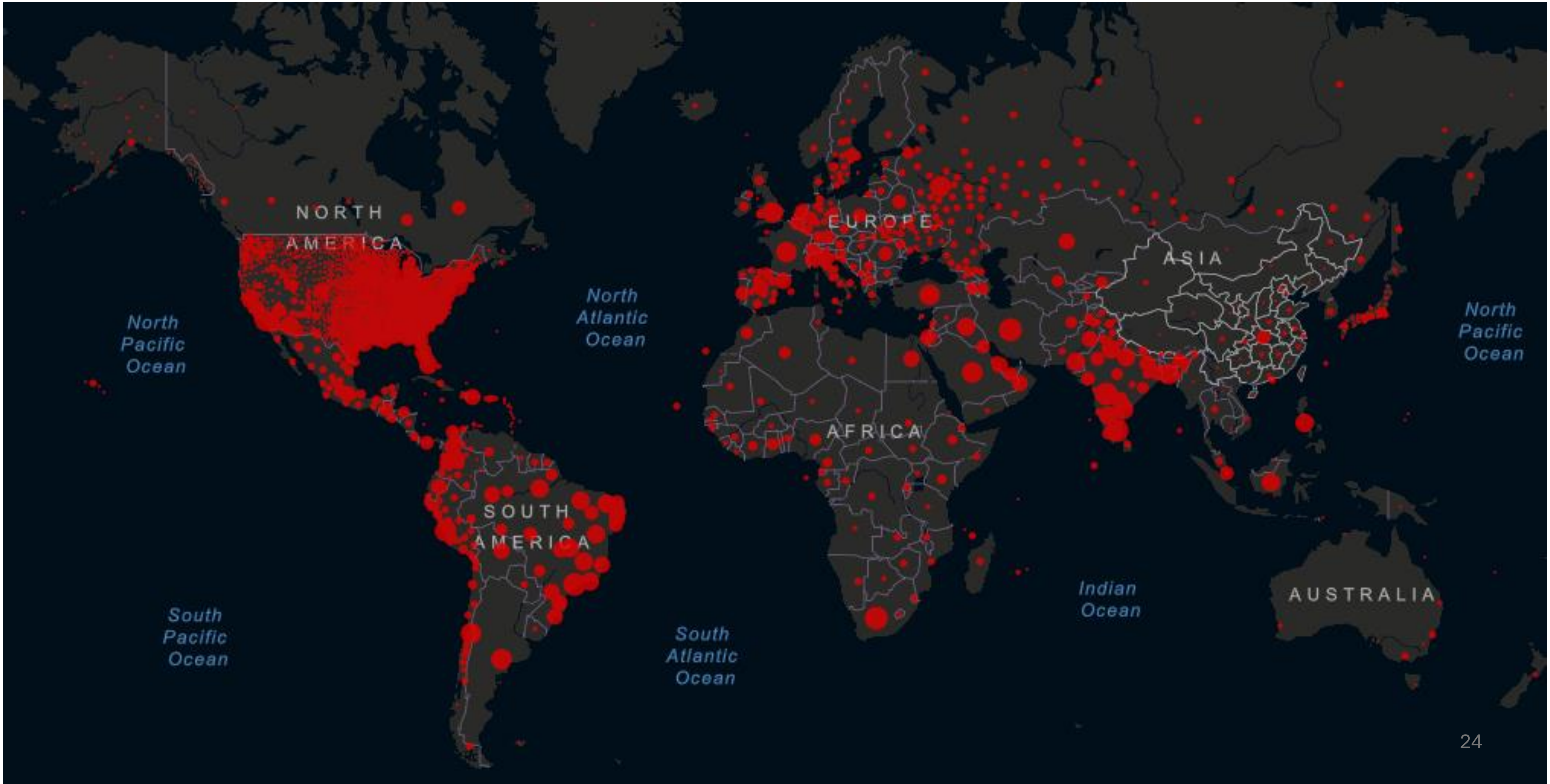




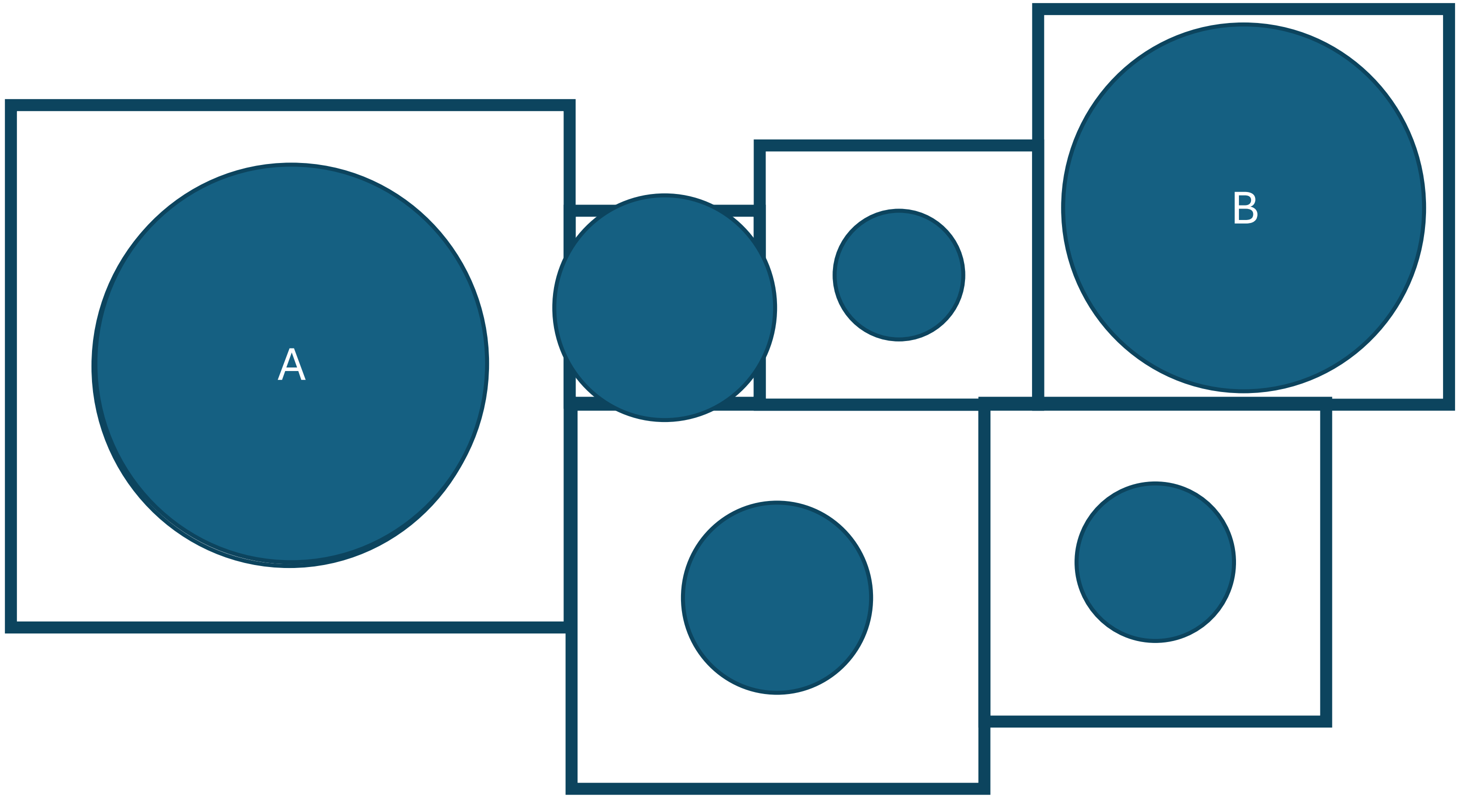
Designing Maps

Maps imply patterns in a spatial context

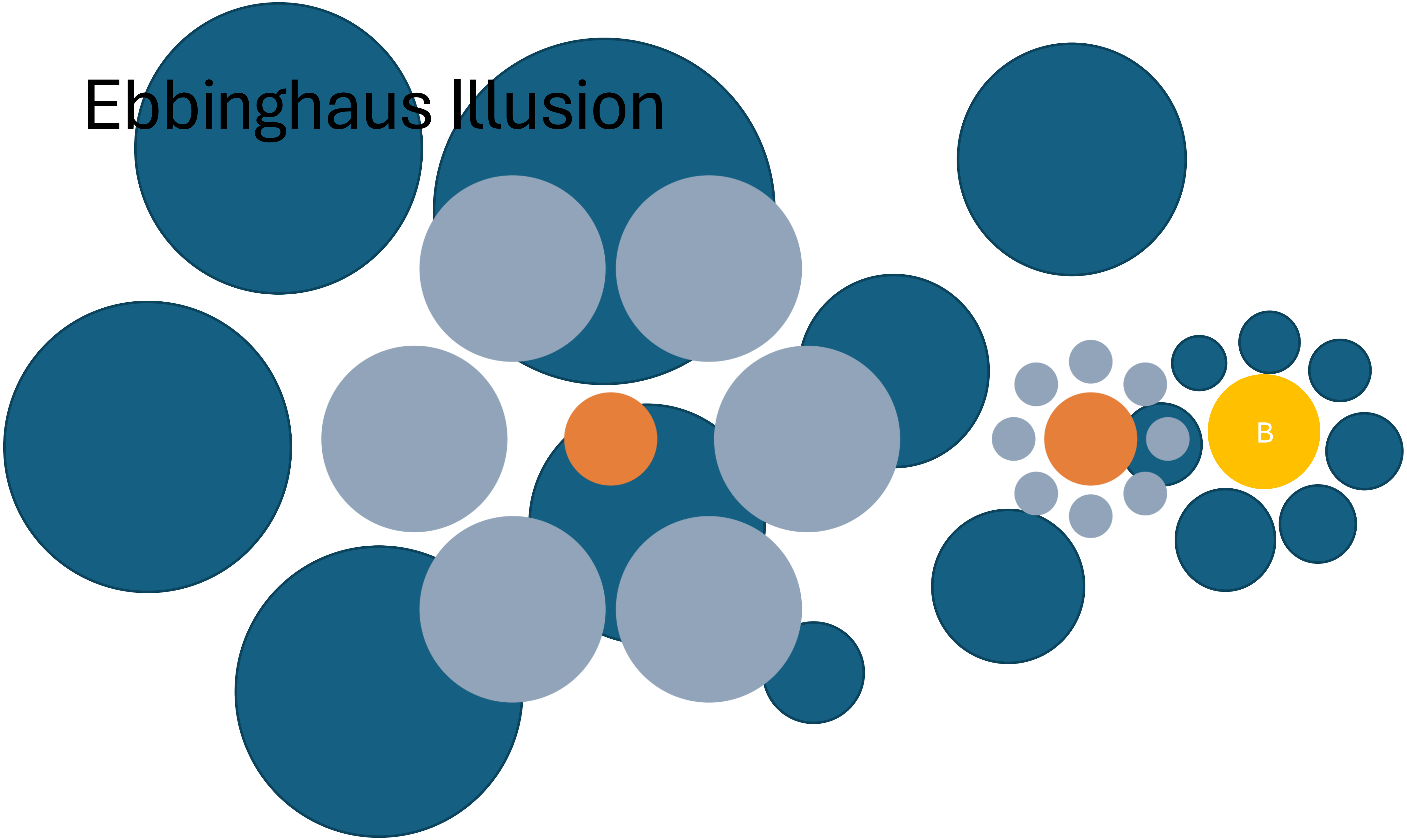
Design Decision: Circles



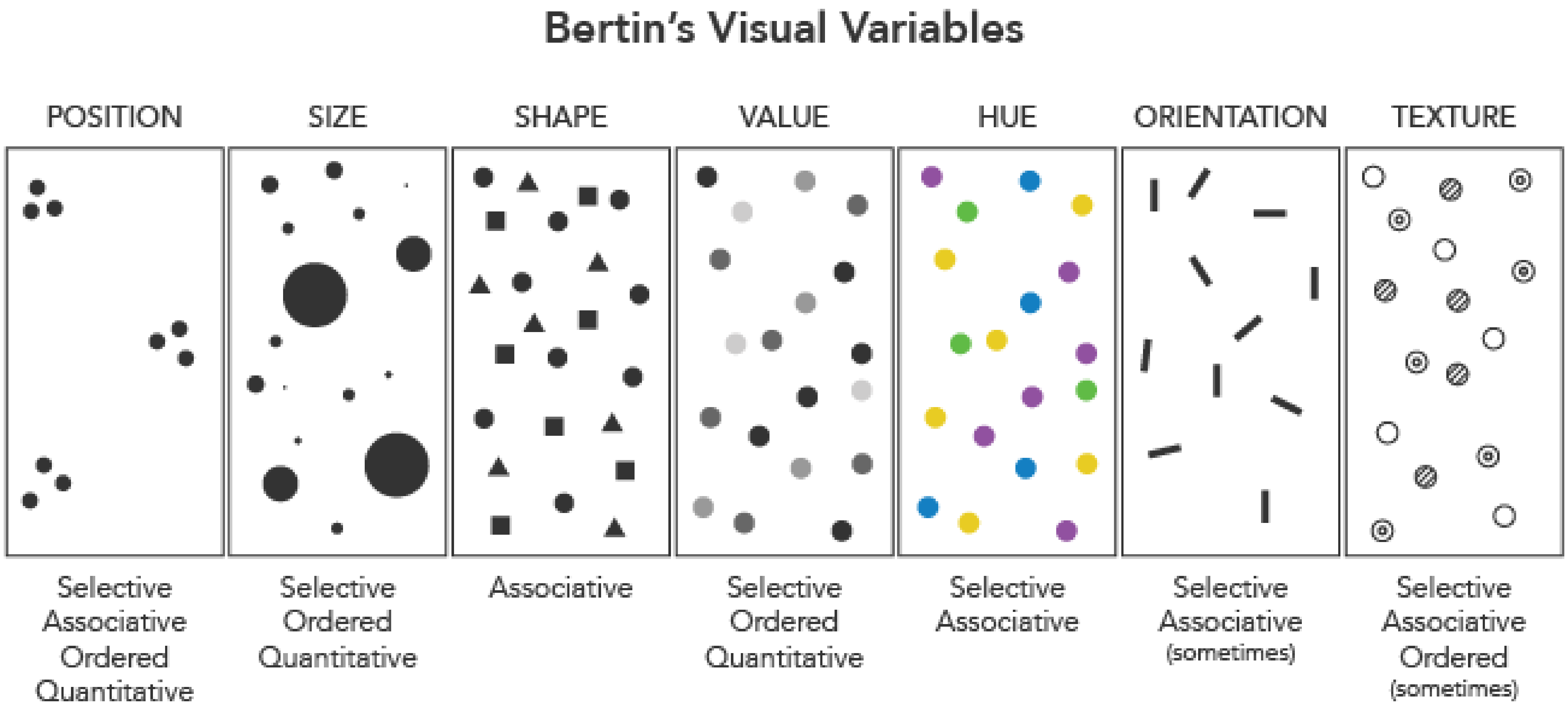
Design Decision: Circles



Ebbinghaus Illusion

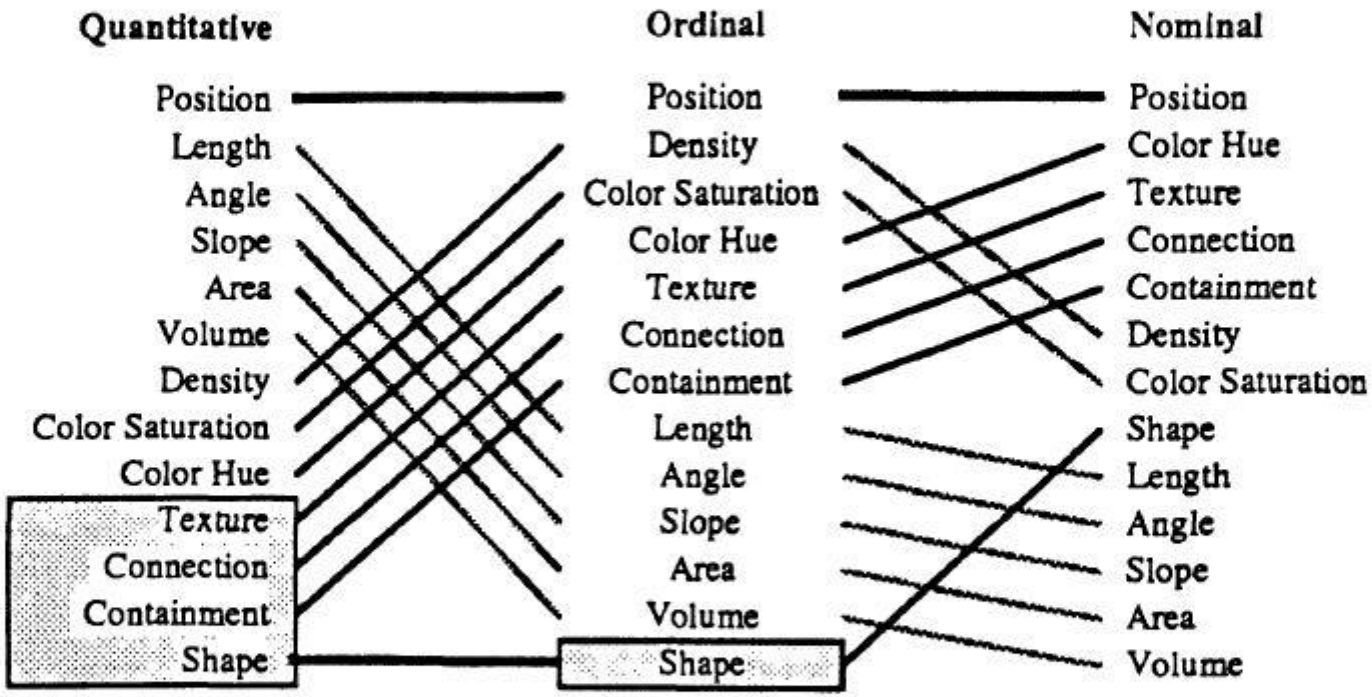


Visualize how?



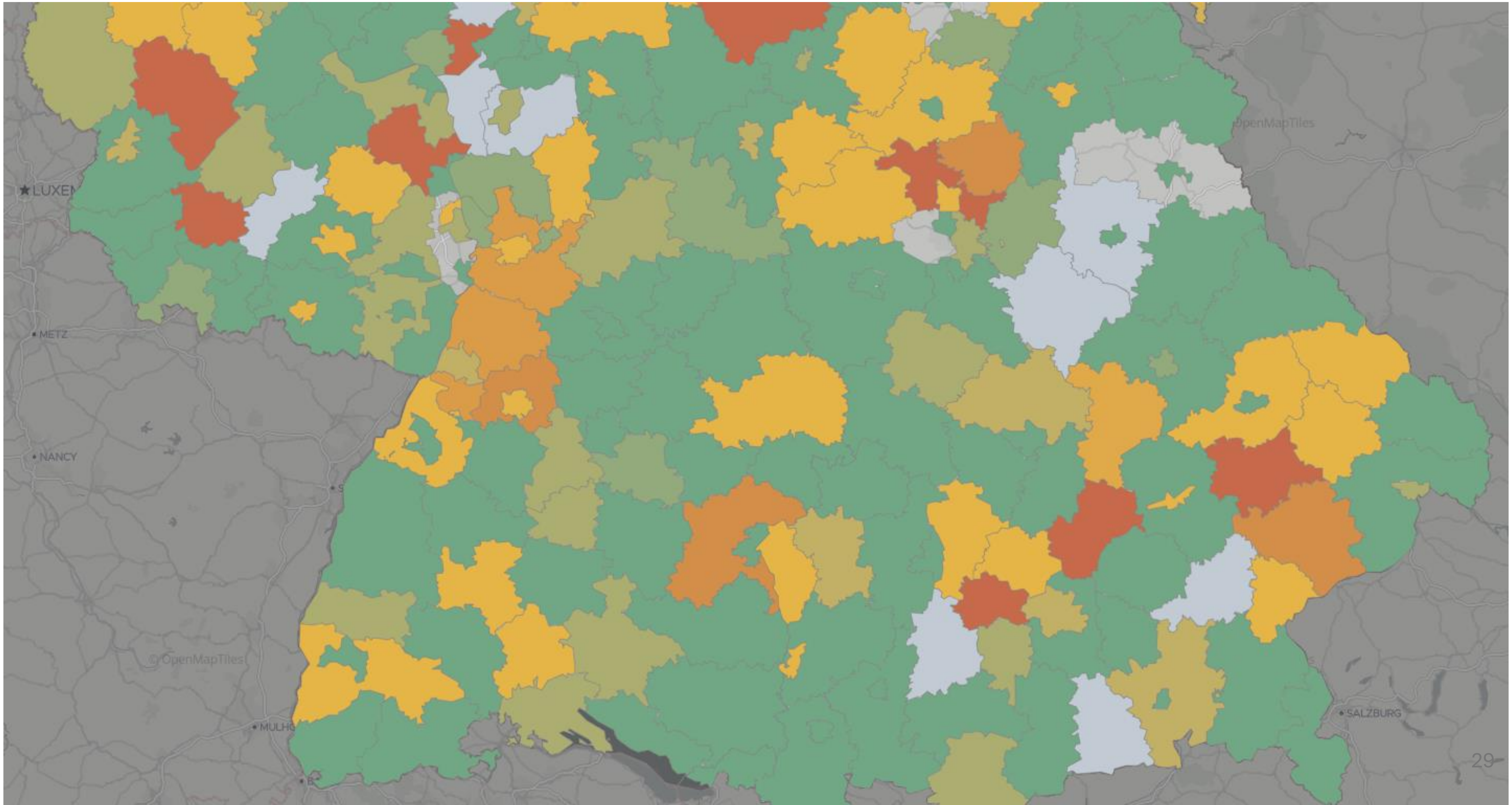
Jaques Bertin, 1974: **Semiology of graphics**

Visual Variables

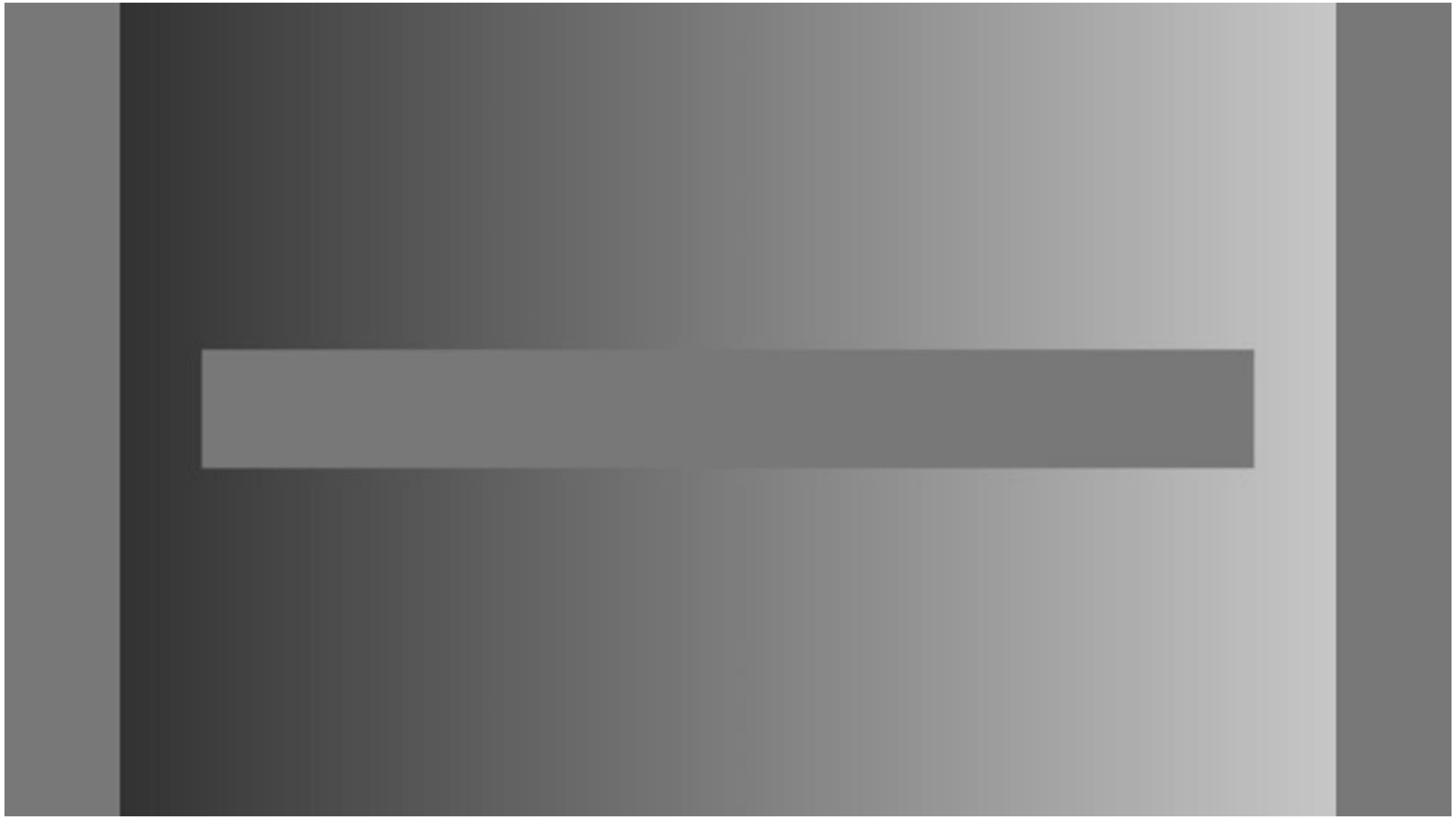


Jock D. Mackinlay, 1986: **Automating the design of graphical presentations of relational information**

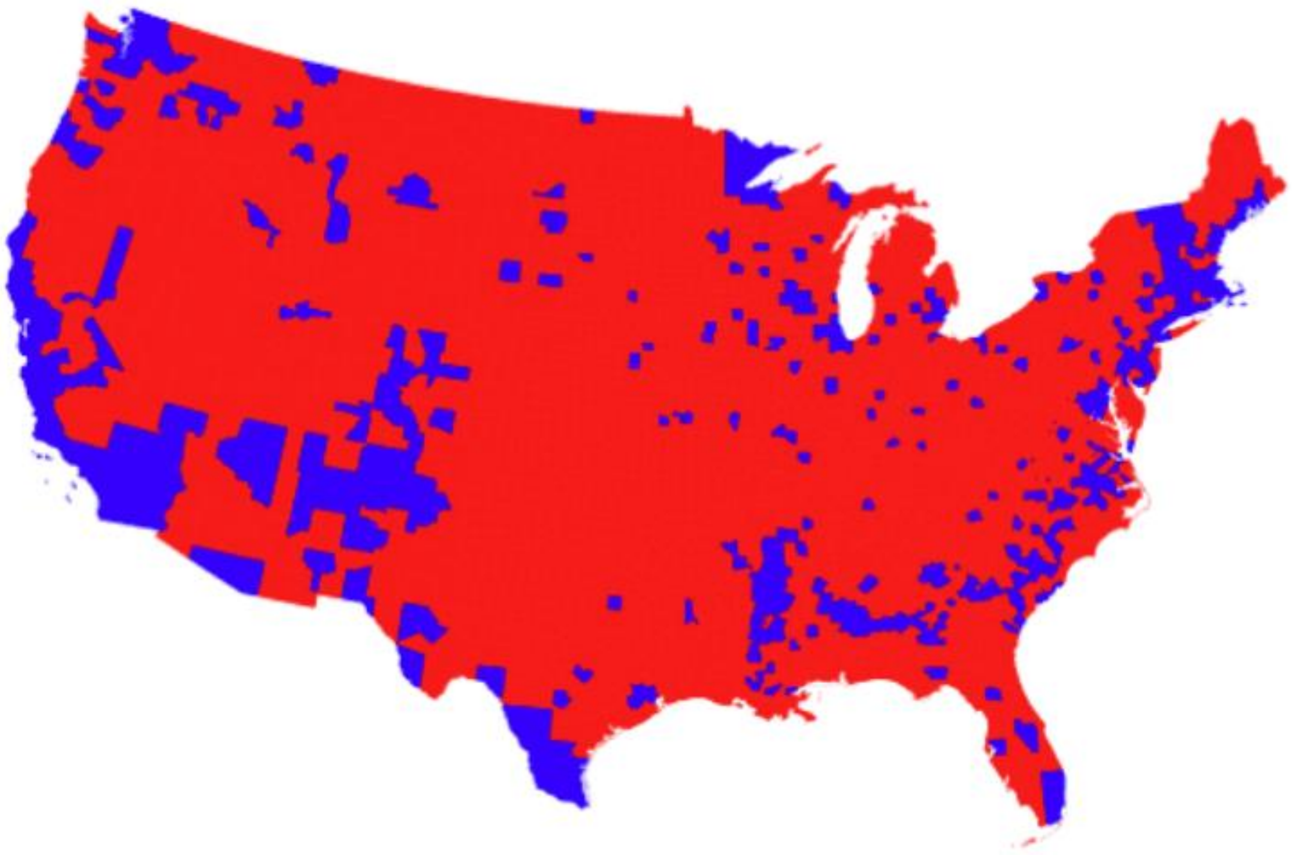
Design Decision: Choropleth Map

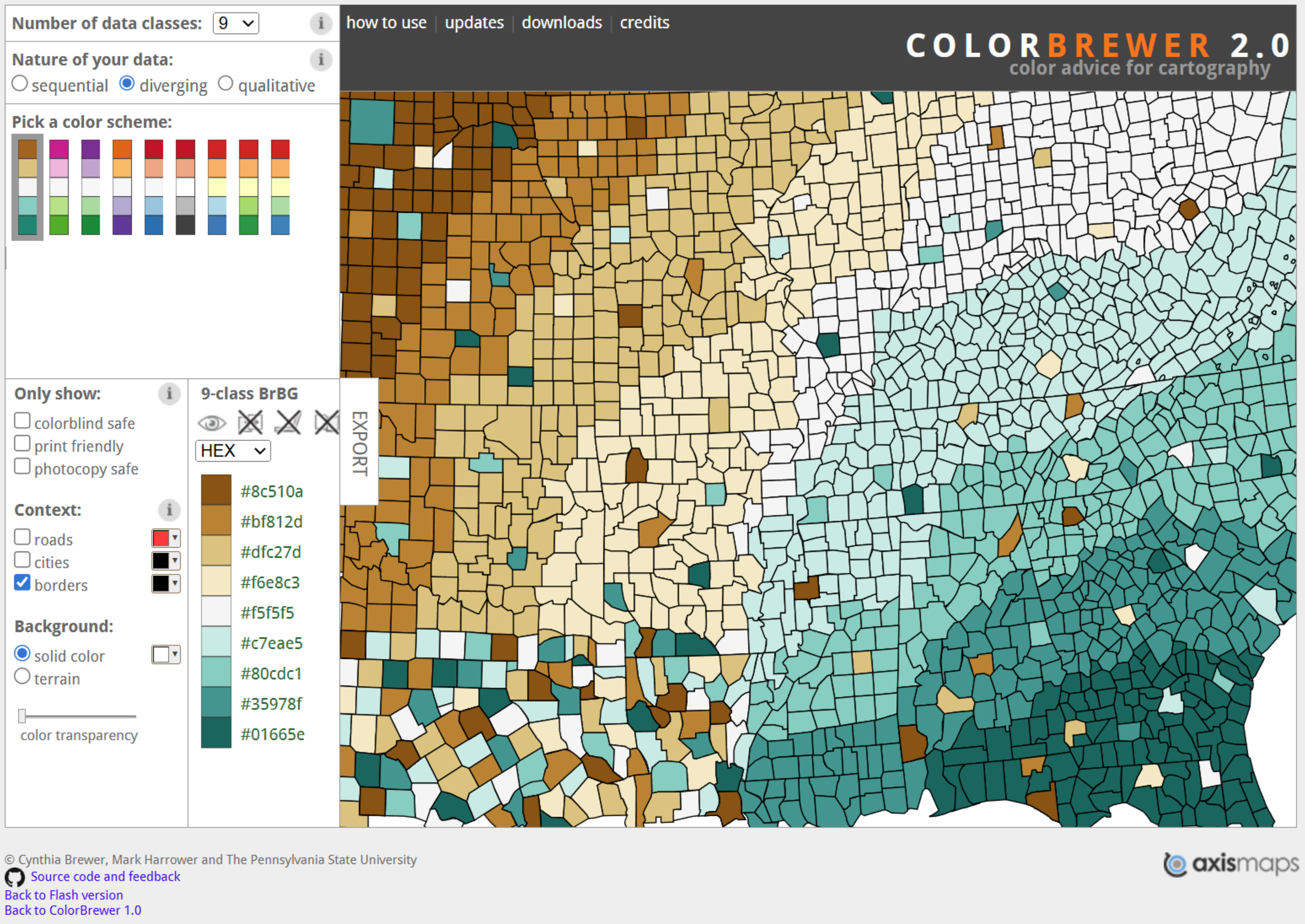


Simultaneous contrast illusions

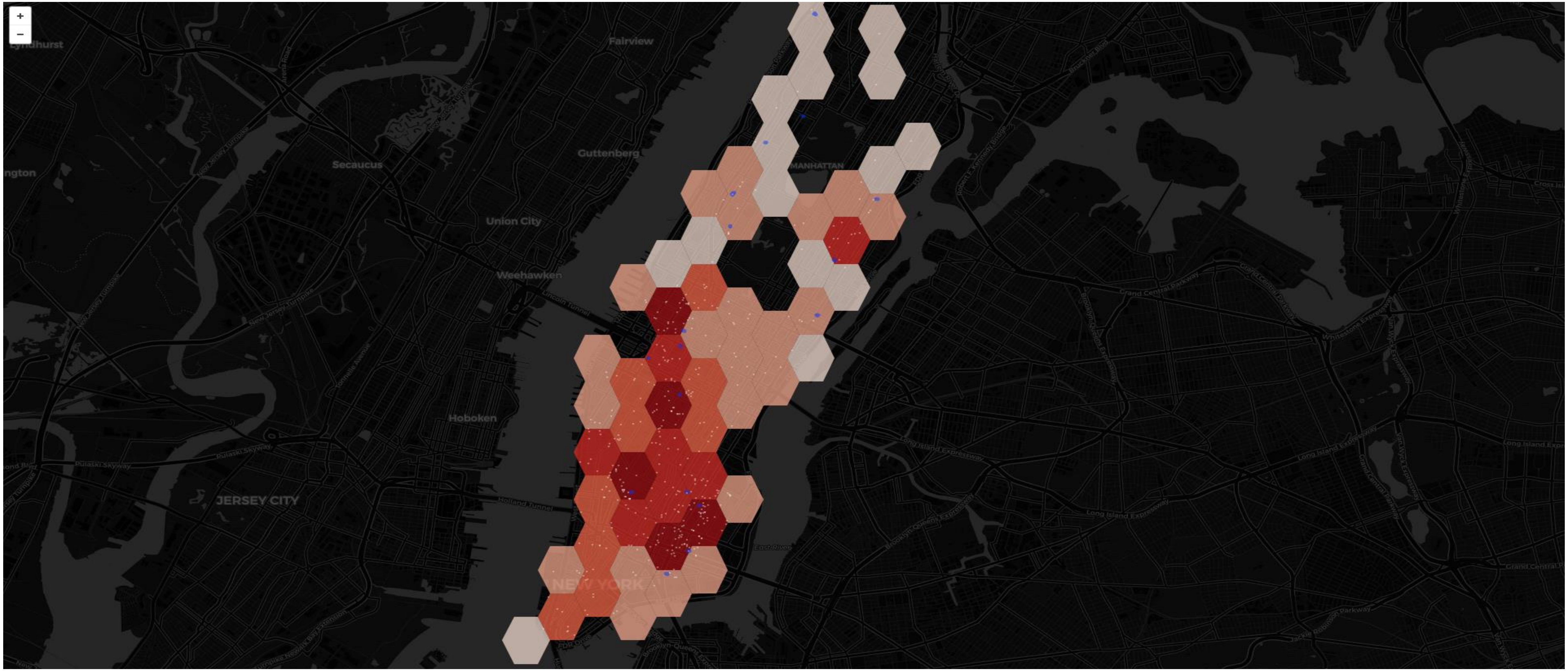


Consider this choropleth map of voters during the United States presidential election in 2016. Geographic regions are broken into counties. Red counties voted majority-Republican; blue counties voted majority-Democratic.

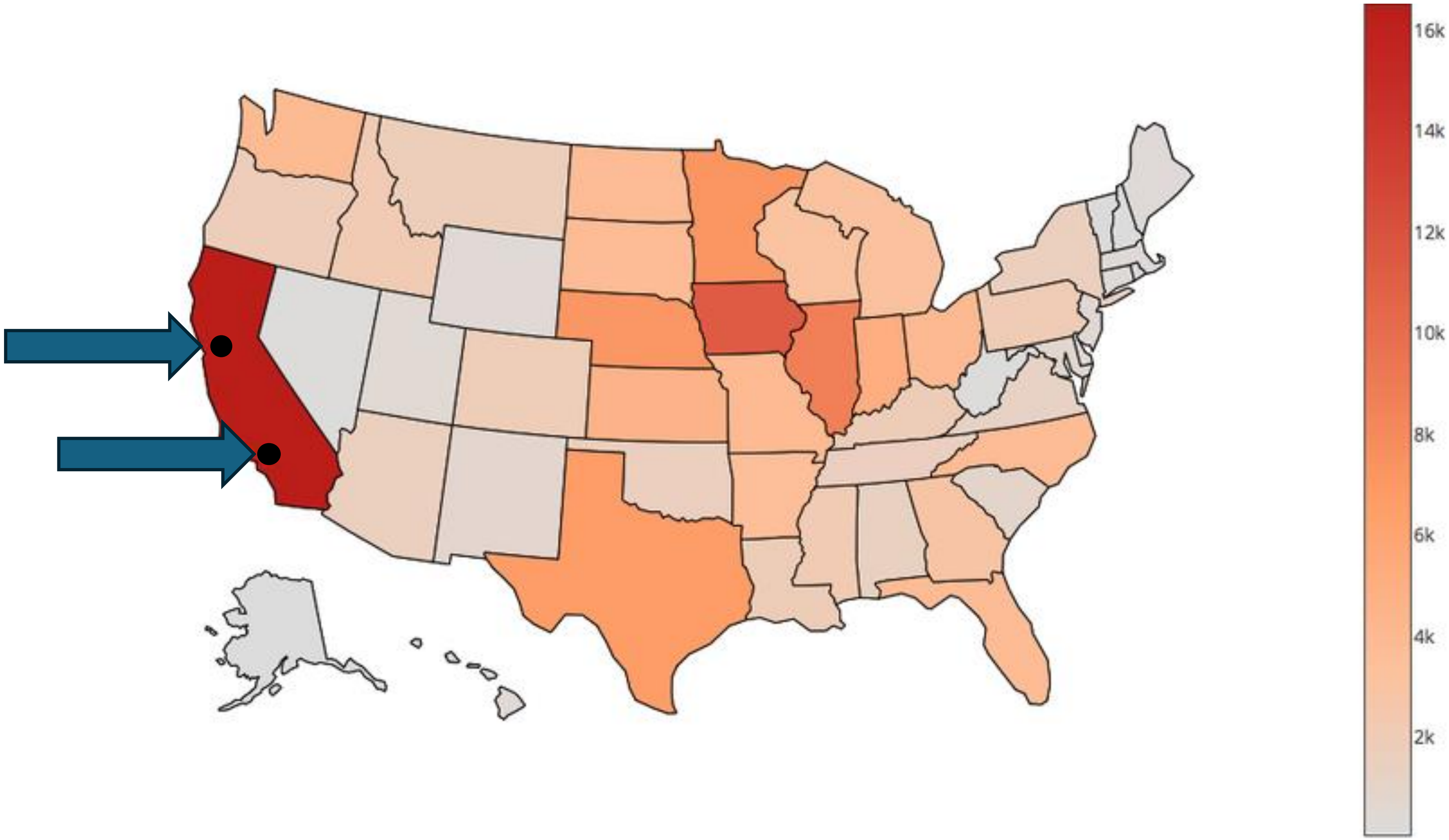



Number of data classes: 9
Nature of your data:
sequential
diverging
qualitative
Pick a color scheme:
how to use | updates | downloads | credits
COLORBREWER 2.0
color advice for cartography
Only show:
colorblind safe
print friendly
photocopy safe
Context:
roads
cities
borders
Background:
solid color
terrain
color transparency
9-class BrBG
HEX
#8c510a
#bf812d
#dfc27d
#f6e8c3
#f5f5f5
#c7eae5
#80cdc1
#35978f
#01665e
EXPORT
© Cynthia Brewer, Mark Harrower and The Pennsylvania State University
Source code and feedback
Back to Flash version
Back to ColorBrewer 1.0
axismaps

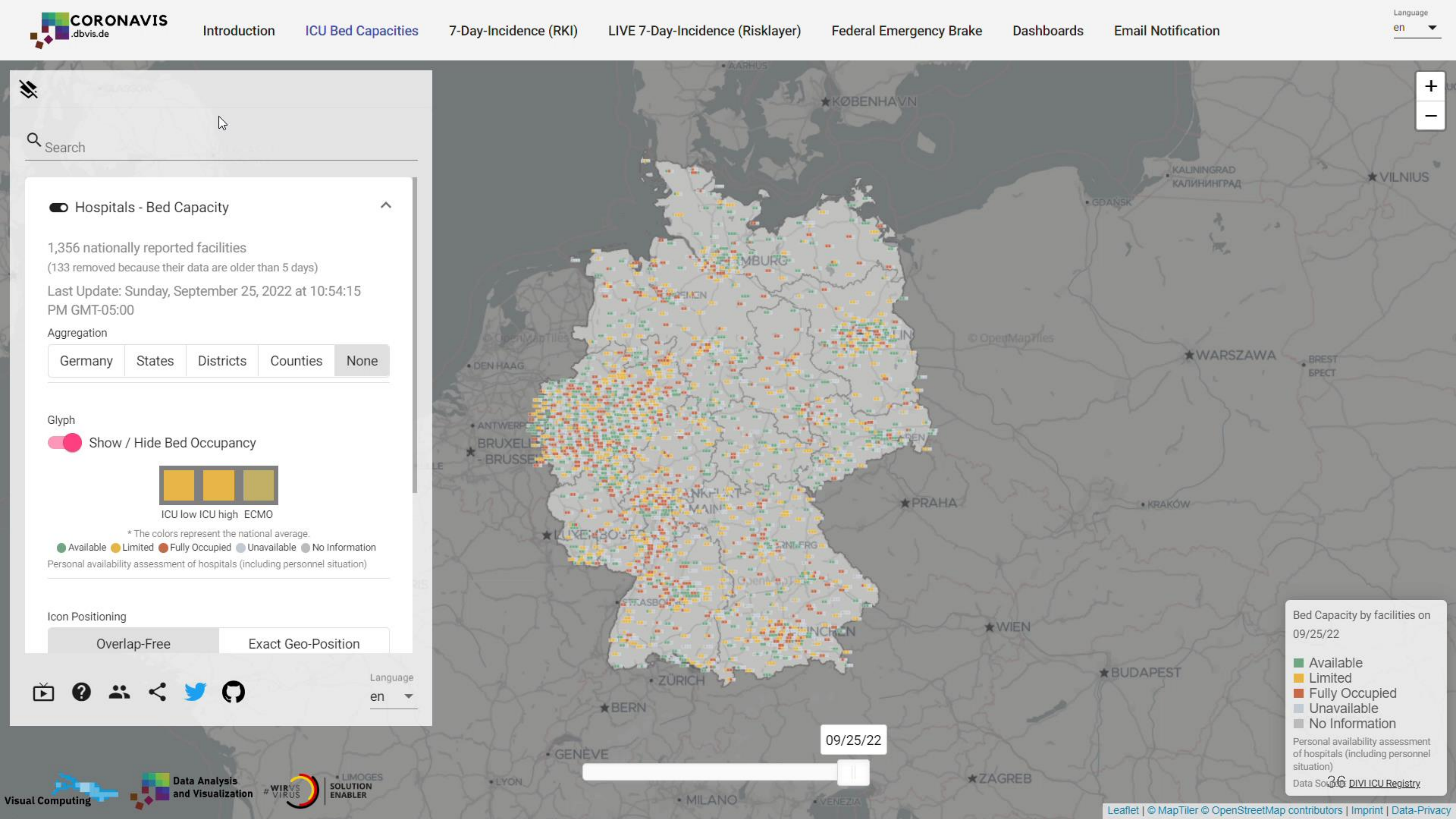
Design Decision: Hexagon Bins



Do not interpolate too much!



Examples: CoronaVis


CORONAVIS .dbvis.de
Introduction
ICU Bed Capacities
7-Day-Incidence (RKI)
LIVE 7-Day-Incidence (Risklayer)
Federal Emergency Brake
Dashboards
Email Notification
Language
en
Search
Hospitals - Bed Capacity
1,356 nationally reported facilities
(133 removed because their data are older than 5 days)
Last Update: Sunday, September 25, 2022 at 10:54:15 PM GMT-05:00
Aggregation
Germany
States
Districts
Counties
None
Glyph
Show / Hide Bed Occupancy
ICU low ICU high ECMO
* The colors represent the national average.
Available Limited Fully Occupied Unavailable No Information
Personal availability assessment of hospitals (including personnel situation)
Icon Positioning
Overlap-Free
Exact Geo-Position
Language
en
Visual Computing
Data Analysis and Visualization
#WIRVSVIRUS SOLUTION ENABLER
KØBENHAVN
KALININGRAD
КАЛИНИНГРАД
VILNIUS
GDANSK
WARSZAWA
BREST
БРЕСТ
DEN HAAG
ANTWERPEN
BRUXELLES - BRUSSEL
PRAHA
KRAKÓW
WIEN
BUDAPEST
ZÜRICH
BERN
GENÈVE
LYON
MILANO
VENEZIA
ZAGREB
© OpenMapTiles
Bed Capacity by facilities on 09/25/22
Available
Limited
Fully Occupied
Unavailable
No Information
Personal availability assessment of hospitals (including personnel situation)
Data Source: DIVI ICU Registry
09/25/22
Leaflet | © MapTiler © OpenStreetMap contributors | Imprint | Data-Privacy

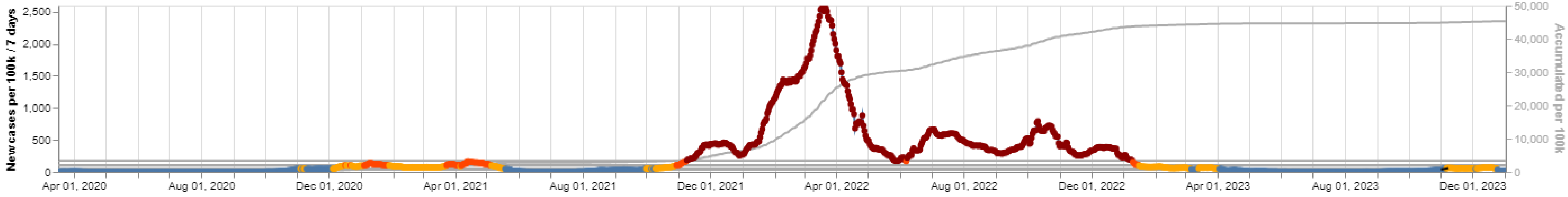
Mecklenburg-Vorpommern

1,608,138 Residents

A non-linear scaling emphasizes low values better.

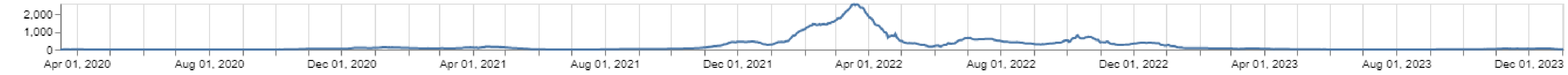
	Current		Change								
	01/03/24, 12:00 AM		Last 24h			Last 72h			last 7d		
Tested positive	727,709	(45,251.65/100k)	0	(0/100k)	(0%)	+2	(+0.12/100k)	(+0%)	+299	**(+18.59/100k)**	(+0%)
Deaths	2,946	(183.19/100k)	0	(0/100k)	(0%)	0	(0/100k)	(0%)	+3	(+0.19/100k)	(+0.1%)
Covid-19 ICU patients	17	(1.06/100k)	-5	(-0.31/100k)	(-22.7%)	-3	(-0.19/100k)	(-15%)	-7	(-0.44/100k)	(-29.2%)
Covid-19 ICU patients (invasive ventilation)	7	(0.44/100k)	-2	(-0.12/100k)	(-22.2%)	-4	(-0.25/100k)	(-36.4%)	-1	(-0.06/100k)	(-12.5%)
Bed occupancy	378/431	(87.7%)	+2/-9		(+2.2%)	+21/+3		(+4.3%)	+18/-7		(+5.5%)

Data Source: Robert Koch Institute (dl-de/by-2-0) data record & Data Source: DIVI ICU Registry



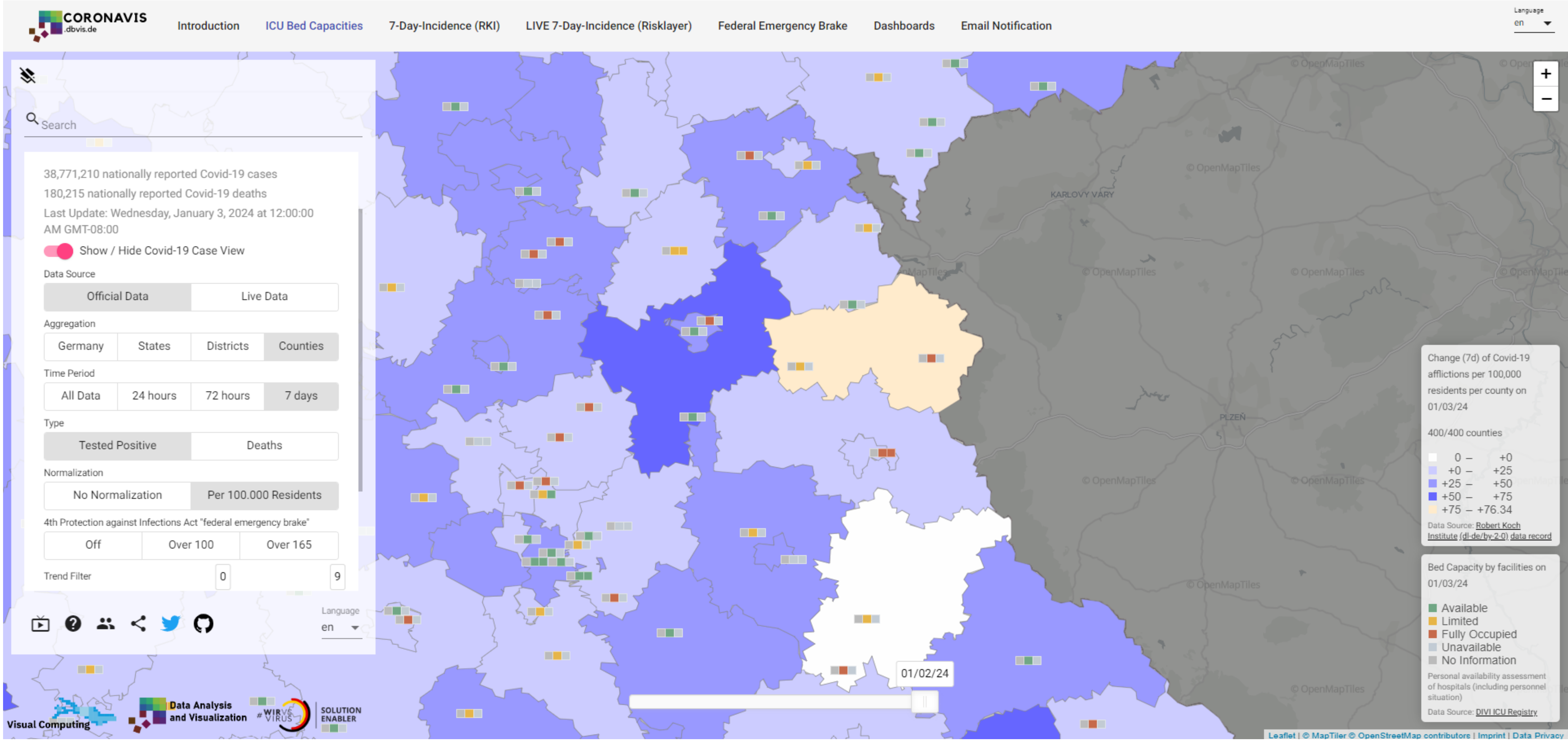
Trend of the last 7 days

+2.9 per day



Drag the mouse here to zoom in on the upper diagram.

Data Source: Robert Koch Institute (dl-de/by-2-0) data record


CORONAVIS
dbvis.de
Introduction
ICU Bed Capacities
7-Day-Incidence (RKI)
LIVE 7-Day-Incidence (Risklayer)
Federal Emergency Brake
Dashboards
Email Notification
Language
en
Search
38,771,210 nationally reported Covid-19 cases
180,215 nationally reported Covid-19 deaths
Last Update: Wednesday, January 3, 2024 at 12:00:00 AM GMT-08:00
Show / Hide Covid-19 Case View
Data Source
Official Data
Live Data
Aggregation
Germany
States
Districts
Counties
Time Period
All Data
24 hours
72 hours
7 days
Type
Tested Positive
Deaths
Normalization
No Normalization
Per 100.000 Residents
4th Protection against Infections Act "federal emergency brake"
Off
Over 100
Over 165
Trend Filter
0
9
Language
en
Visual Computing
Data Analysis and Visualization
#WIRVSVIRUS
SOLUTION ENABLER
KARLOVY VARY
PLZEŇ
© OpenMapTiles
01/02/24
Change (7d) of Covid-19 afflictions per 100,000 residents per county on 01/03/24
400/400 counties
0 – +0
+0 – +25
+25 – +50
+50 – +75
+75 – +76.34
Data Source: Robert Koch Institute (dl-de/by-2-0) data record
Bed Capacity by facilities on 01/03/24
Available
Limited
Fully Occupied
Unavailable
No Information
Personal availability assessment of hospitals (including personnel situation)
Data Source: DIVI ICU Registry
Leaflet | © MapTiler © OpenStreetMap contributors | Imprint | Data Privacy

CORONAVIS .dbvis.de

Universität Konstanz

Configurable Dashboards

WHO COVID-19 dashboard

WHO Health Emergencies Programme

World

Summary | Circulation | Cases | Deaths | Hospitalizations | Vaccines | More

COVID-19 Cases, World

WHO will utilize various sources to continue monitoring the COVID-19 epidemiological situation via the WHO COVID-19 dashboard. If data for certain countries is unavailable in this section, it may indicate that they have either ceased reporting COVID-19 surveillance data to WHO or have integrated the COVID-19 surveillance into existing respiratory disease surveillance. In the latter case, SARS-CoV-2 detections from sentinel and systematic virological surveillance sites in those countries may be found in the Circulation section which also includes information on SARS-CoV-2 variant circulation. This global summary of COVID-19 cases includes data on confirmed cases reported to WHO from the comprehensive COVID-19 case monitoring.

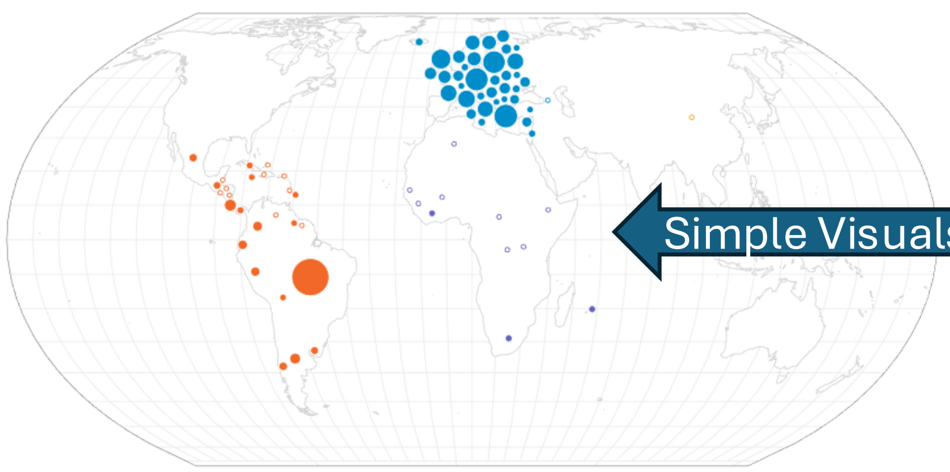
Explanation

Last 7 days | Last 28 days | Total cumulative

Count | Rate per 100 000

Number of COVID-19 cases reported to WHO

World, 28 days to 28 December 2025



Simple Visuals

WHO Regions: Africa, Americas, Eastern Mediterranean, Europe, South-East Asia, Western Pacific

57,813 −18,333 decrease on previous 28 days

Context

Reported COVID-19 cases

World, 28 days to 28 December 2025

Main Point

Number of COVID-19 cases reported to WHO

World, 28 days to 28 December 2025

Country	Cases
World	57.8k
Brazil	18.3k
Greece	5.6k
Show 71 more	
Suriname	0
Turks And Caicos Islands	0
Venezuela (Bolivarian Republic of)	0

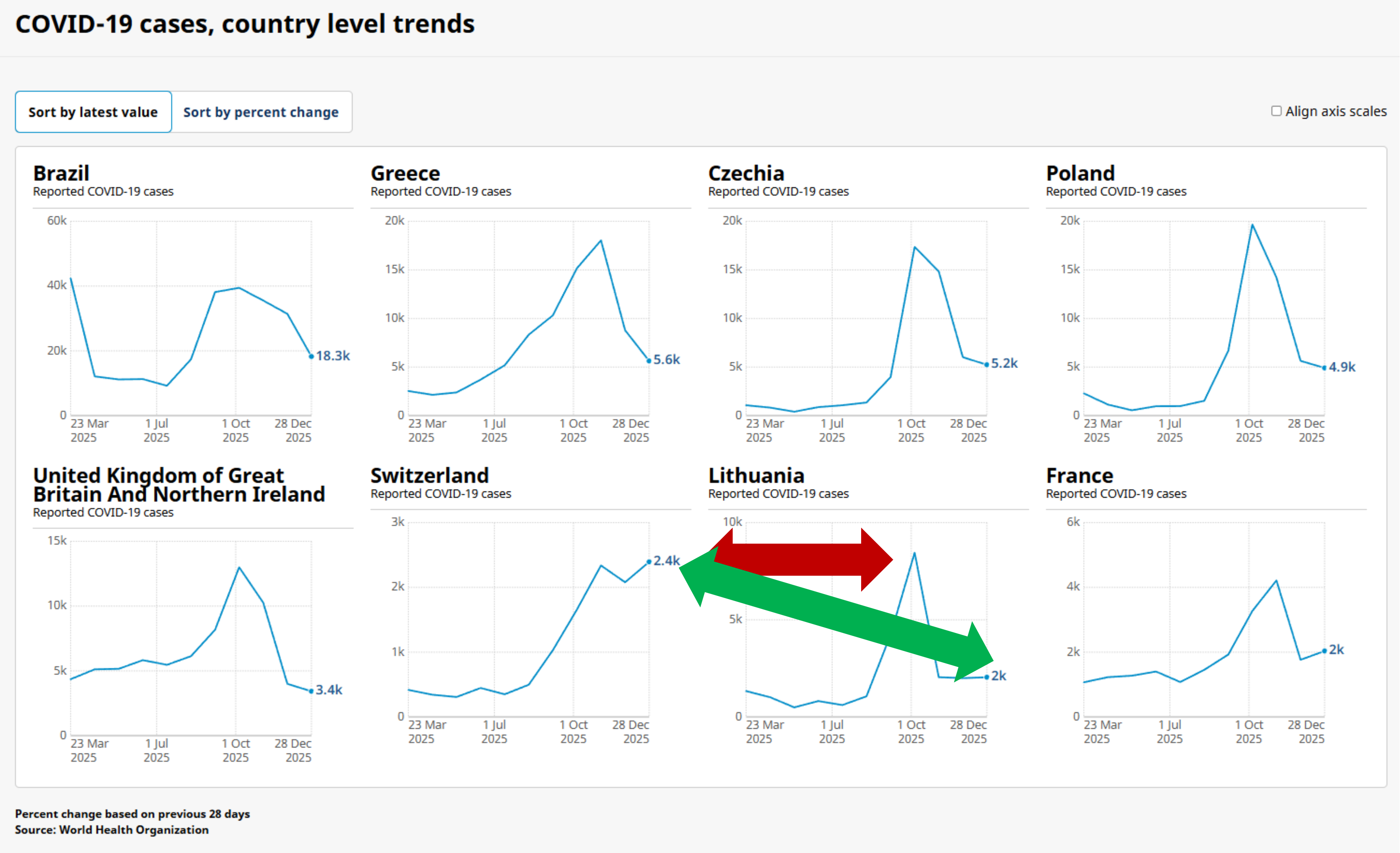
Clear Overview

Source: World Health Organization

Most recent data submission date: 27th December 2025

Data Updated

Careful!



COVID-19 cases, country level trends
Sort by latest value
Sort by percent change
Align axis scales
Brazil
Reported COVID-19 cases
60k
40k
20k
0
18.3k
Greece
Reported COVID-19 cases
20k
15k
10k
5k
0
5.6k
Czechia
Reported COVID-19 cases
20k
15k
10k
5k
0
5.2k
Poland
Reported COVID-19 cases
20k
15k
10k
5k
0
4.9k
United Kingdom of Great Britain And Northern Ireland
Reported COVID-19 cases
15k
10k
5k
0
3.4k
Switzerland
Reported COVID-19 cases
3k
2k
1k
0
2.4k
Lithuania
Reported COVID-19 cases
10k
5k
0
2k
France
Reported COVID-19 cases
6k
4k
2k
0
2k
23 Mar 2025
1 Jul 2025
1 Oct 2025
28 Dec 2025
Percent change based on previous 28 days
Source: World Health Organization

Visualization Foundations for Public Health

Thank you!